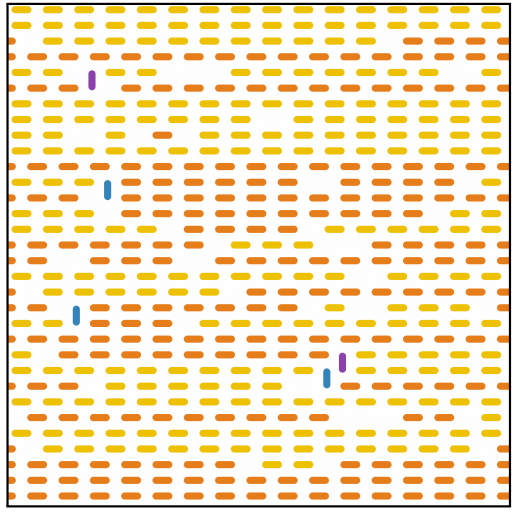
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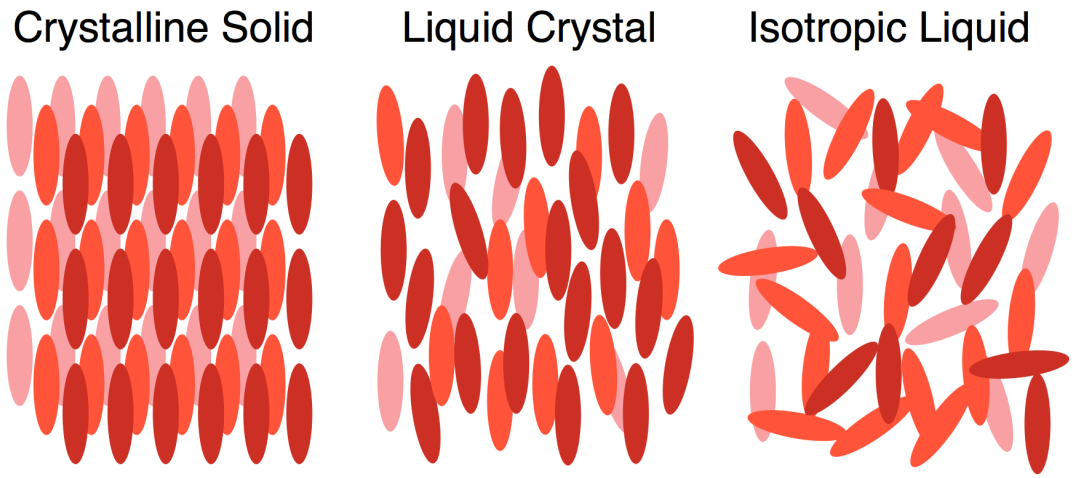
Nematic liquid-crystalline phase in an interacting monomer-dimer model of Heilmann and Lieb

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Liquid crystals



- A state of matter intermediate between a liquid and a solid
- Exhibit anisotropic electrical, magnetic, and optical properties
- Used in liquid-crystal displays (LCDs)
- Characterized by the presence of orientational order and a partial or complete absence of positional order

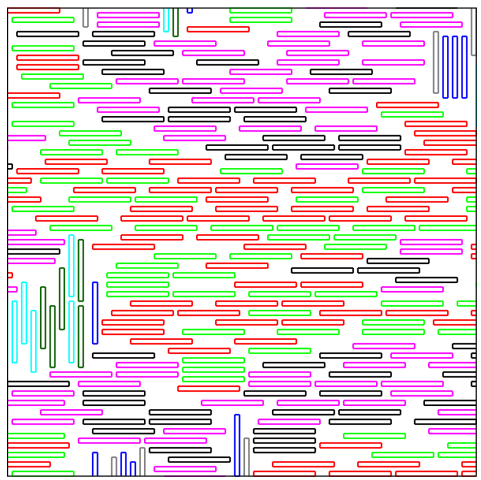
Question: Can we rigorously prove that a microscopic model exhibits a liquid-crystalline phase?

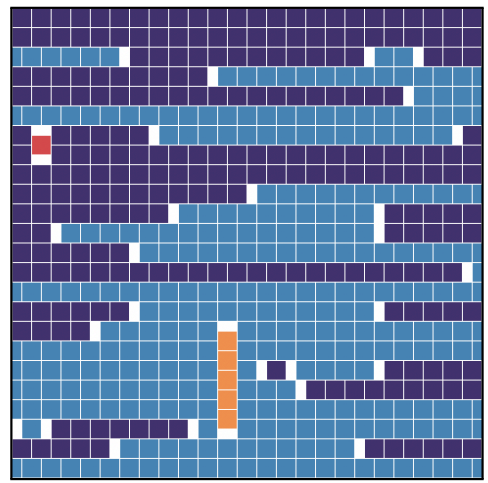
Mathematical results on liquid crystals

This has been achieved only in a small number of models, including:

- **Disertori–Giuliani (CMP, 2013):** long rods on $\mathbb{Z}^2$, two orientations
  - **H. (2026):** long rectangles on $\mathbb{Z}^2$, with explicit estimates of the required aspect ratio ($k \geq 10^{72}$; simulations suggest $k_{\min} = 7$)
- **Disertori–Giuliani–Jauslin (CMP, 2020):** anisotropic plates in $\mathbb{R}^3$, six orientations
- **Hadas–Peled (JEMS, 2026):** 2×2 squares on $\mathbb{Z}^2$

In all of the above, the only interaction between particles is a geometric exclusion forbidding overlaps

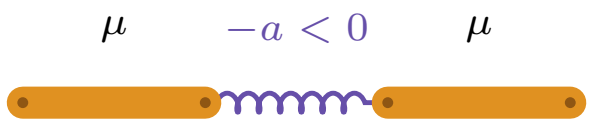




An interacting monomer-dimer model

Model I of Heilmann–Lieb (JSP, 1979)

- Non-intersecting dimers (edges) on the square lattice $\mathbb{Z}^2$
  - Vertices not covered by a dimer are the monomers
- Dimers are at chemical potential μ
- An attractive energy $-a < 0$ is associated with each pair of adjacent and collinear dimers



The **Hamiltonian** (total energy) of a dimer configuration σ is

$$H(\sigma) = -\mu|\sigma| - a \sum_{\{e,e'\}\subset\sigma} \mathbb{1}_{e,e' \text{ adjacent and collinear}}.$$

Statistical mechanics tells us that the probability of observing σ at inverse temperature β is given by the **Gibbs measure**

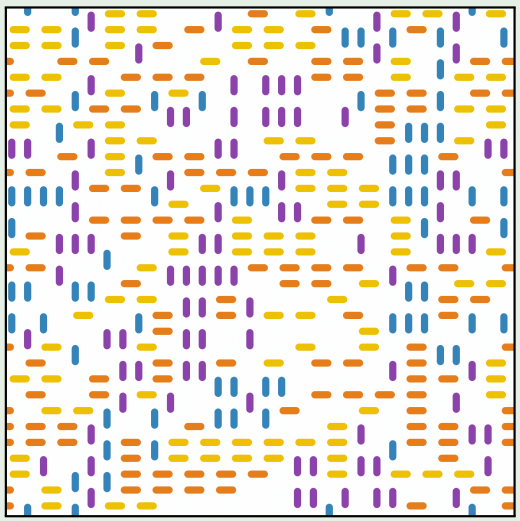
$$\mathbb{P}_\beta(\sigma) \propto e^{-\beta H(\sigma)}.$$

The Heilmann–Lieb conjecture on Model I

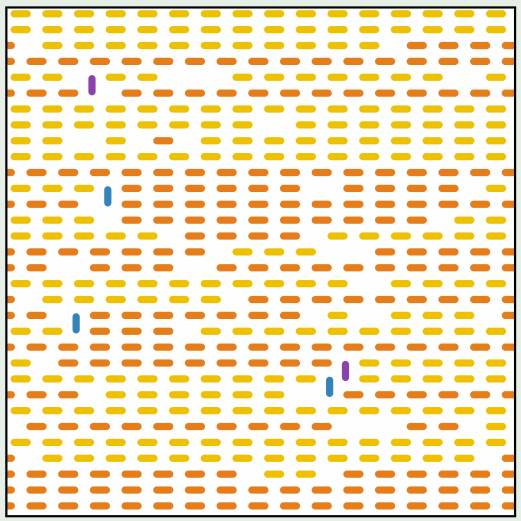
Conjecture (Heilmann–Lieb, 1979)

Suppose that $\mu + a > 0$. At sufficiently low temperatures ($\beta \gg 1$), the model exhibits nematic order: a typical sample from a Gibbs measure $\mathbb{P}_\beta$ is orientationally ordered but positionally disordered.

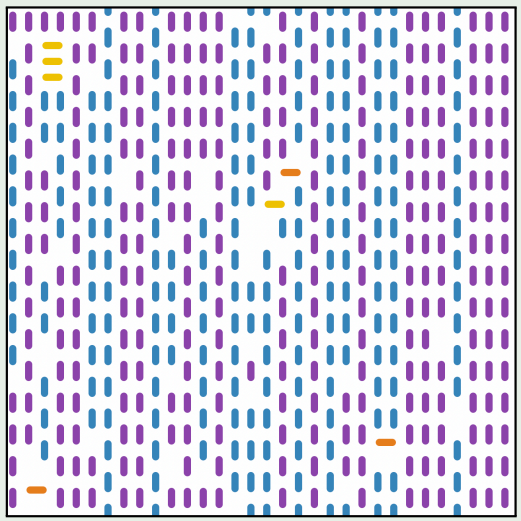
Snapshots from a Monte Carlo simulation



$\mu = 0.1, a = 0.3, \beta = 3.0$



$\beta = 5.6$



$\beta = 5.6$

Mathematical results on Model I

Conjecture (Heilmann–Lieb, 1979)

Nematic order at $\beta \gg 1$ when $\mu + a > 0$.

Theorem (Heilmann–Lieb, 1979)

Orientational order at $\beta \gg 1$ when $\mu + a > 0$.

Theorem (Alberici, JSP, 2016)

Unequal chemical potentials for horizontal and vertical dimers.
Nematic order at $\beta \gg 1$ when $\mu_{\mathrm{hor}} + a > 0$ and $2\mu_{\mathrm{ver}} + 5a < 0$.

Theorem (Jauslin–Lieb, CMP, 2018)

Nematic order when $\beta a \gg e^{\beta\mu} \gg 1$.

Theorem (H., 2025)

Nematic order at $\beta \gg 1$ when $\mu + a > 0$ and $3a > \mu$.

Main results

Theorem (H., 2025)

Assume $\mu + a > 0$ and $3a > \mu$. There exist $\beta_0, C, c > 0$ and $\mathfrak{n} \geq 100$ s.t., for all $\beta \geq \beta_0$, Model I admits a unique Gibbs measure $\mathbb{P}_{\mathrm{ver}}$ s.t.:

1. *$\mathbb{P}_{\mathrm{ver}}$ is $\mathbb{Z}^2$-translation invariant and extremal.*
2. *For all edges e of $\mathbb{Z}^2$,*

$$\mathbb{P}_{\mathrm{ver}}(\text{dimer at } e) = \begin{cases} \frac{1}{2} - \mathcal{O}\left(e^{-\frac{1}{2}\beta(\mu+a)}\right) & \text{if } e \text{ is vertical} \\ \mathcal{O}\left(e^{-3\beta(1-1/\mathfrak{n})a}\right) & \text{if } e \text{ is horizontal} \end{cases}.$$

3. *For all edges e, e' of $\mathbb{Z}^2$,*

$$\begin{aligned} &|\mathrm{Cov}_{\mathbb{P}_{\mathrm{ver}}}(\text{dimer at } e, \text{ dimer at } e')| \\ &\qquad \leq C \exp\left\{-c\left[|x_e - x_{e'}| + e^{-\frac{1}{2}\beta(\mu+3a)}|y_e - y_{e'}|\right]\right\}. \end{aligned}$$

The push-forward measure $\mathbb{P}_{\mathrm{hor}} = \tau\mathbb{P}_{\mathrm{ver}}$, where τ is the reflection $(x, y) \mapsto (y, x)$, satisfies analogous properties. Moreover, every periodic Gibbs measure is a convex combination of $\mathbb{P}_{\mathrm{ver}}$ and $\mathbb{P}_{\mathrm{hor}}$.

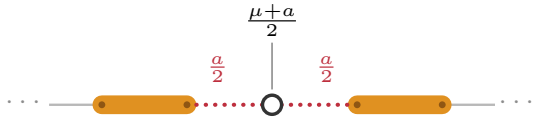
Physical interpretations of the conditions

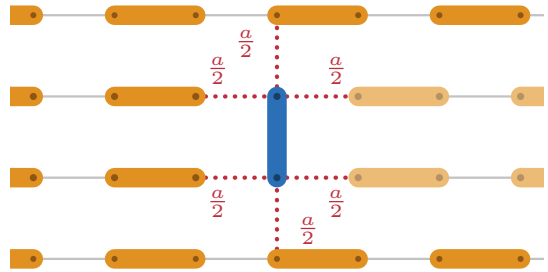
The Hamiltonian is equivalently rewritten as

$$H_{\text{shifted}}(\sigma) = \frac{\mu + a}{2} \#\text{vacancies} + \frac{a}{2} \#\text{broken links}$$



1. $a > 0$: built into the model, favors alignment
2. $\mu + a > 0$: favors dense configurations
3. In an otherwise perfectly aligned environment, an isolated vacancy is energetically cheaper than a misaligned dimer precisely when $3a > \mu$.





Separation of scales: misalignment is rare on the length scale on which we expect to find vacancies.

Strongly percolating sets

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A **random set** is a measurable function $\Omega \to 2^{\mathbb{Z}^2}$. Let $\epsilon \in [0, 1/21)$.

Definition (ϵ-rarity)

A random set $B \subset \mathbb{Z}^2$ is **ϵ-rare** if, for all finite $A \subset \mathbb{Z}^2$,

$$\mathbb{P}(A \subset B) \leq \epsilon^{|A|}.$$

Definition (ϵ-strong percolation)

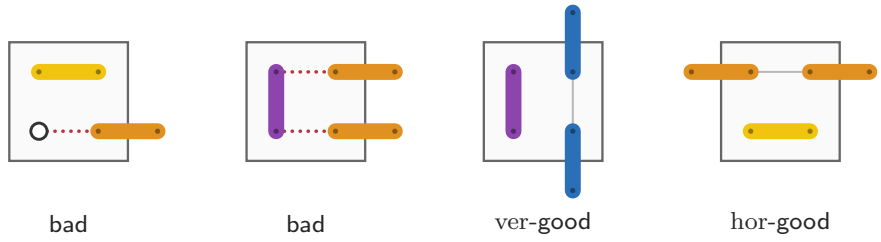
A random set $\Psi \subset \mathbb{Z}^2$ is **ϵ-strongly percolating** if there exists an ϵ-rare random set $B \subset \mathbb{Z}^2$ s.t. Ψ contains the unique infinite component of $\mathbb{Z}^2 \setminus B$ a.s. In this case, we write

$$\mathfrak{p}_{\mathbb{P}}(\Psi) \leq \epsilon.$$

1. ϵ-rarity measures how much it hurts for B to be big.
2. ϵ-strong percolation measures how much it hurts not to be in Ψ.
3. The smaller ϵ is, the more it hurts.

Step 1: Heilmann–Lieb's characterization of orientational order

Declare a 2×2 square R **bad** if it *contains* a vacancy or it intersects dimers of both orientations (and so *contains* a broken link).



$$\|R \text{ is bad}\|_R \leq Ce^{-\beta \min\{\mu+a,a\}/2}.$$

For $o \in \{\mathrm{ver}, \mathrm{hor}\}$, let $\Psi_o \subset \mathbb{Z}^2$ be the random set consisting of the lower-left corners (up to a shift of $(-\frac{1}{2}, -\frac{1}{2})$) of all o-good squares.

1. $\mathrm{d}_\infty(\Psi_{\mathrm{ver}}, \Psi_{\mathrm{hor}}) \geq 2$,
2. $\mathfrak{p}_{\mathbb{P}}(\Psi_{\mathrm{ver}} \cup \Psi_{\mathrm{hor}}) \leq \|R \text{ is bad}\|_R^{1/4}$,
3. $\min\{\mathfrak{p}_{\mathbb{P}}(\Psi_{\mathrm{ver}}), \mathfrak{p}_{\mathbb{P}}(\Psi_{\mathrm{hor}})\} \leq \mathfrak{p}_{\mathbb{P}}(\Psi_{\mathrm{ver}} \cup \Psi_{\mathrm{hor}})$.

Proposition

If $\beta \gg 1$, with respect to any $\mathbb{Z}^2$-ergodic Gibbs measure $\mathbb{P}$, exactly one of Ψ_o percolates, while the other has finite components a.s.

Step 2: Microscopic rarity of misaligned dimers (1/3)

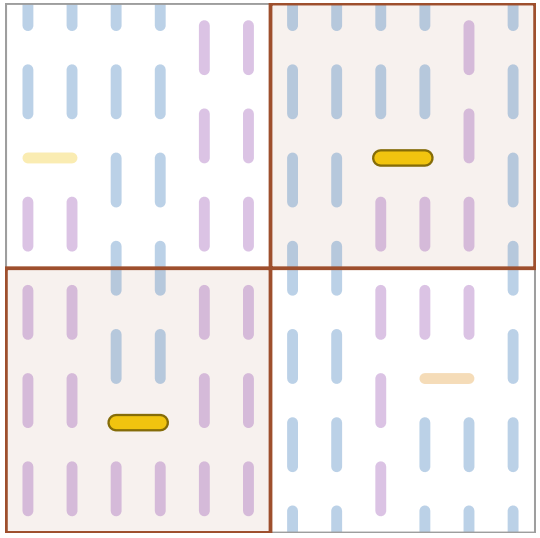
Let $\mathbb{P}_{\mathrm{ver}}$ be a $\mathbb{Z}^2$-ergodic Gibbs measure under which Ψ_{ver} percolates.

Claim

Under $\mathbb{P}_{\mathrm{ver}}$, horizontal dimers are significantly rarer than the bound $\mathfrak{p}_{\mathbb{P}_{\mathrm{ver}}}(\Psi_{\mathrm{ver}}) \leq Ce^{-\beta \min\{\mu+a,a\}/8}$ suggests.

Fix an even $\mathfrak{n} = \mathfrak{n}(\mu, a) \geq 100$. Define the random set

$$\Theta_{\mathrm{ver}} = \big\{(x,y) \in \mathbb{Z}^2 \mid \text{the horizontal edge below the center of the } \mathfrak{n} \times \mathfrak{n} \text{ square centered at } (\mathfrak{n}x, \mathfrak{n}y) \text{ is not covered by a dimer}\big\}.$$



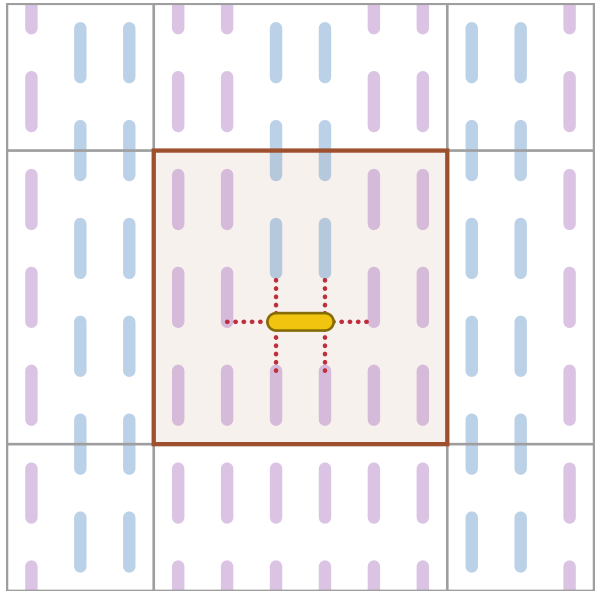
Step 2: Microscopic rarity of misaligned dimers (2/3)

Proposition

If $\beta \gg 1$, then $\mathfrak{p}_{\mathbb{P}_{\mathrm{ver}}}(\Theta_{\mathrm{ver}}) \leq Ce^{-3\beta(1-1/\mathfrak{n})a}$.

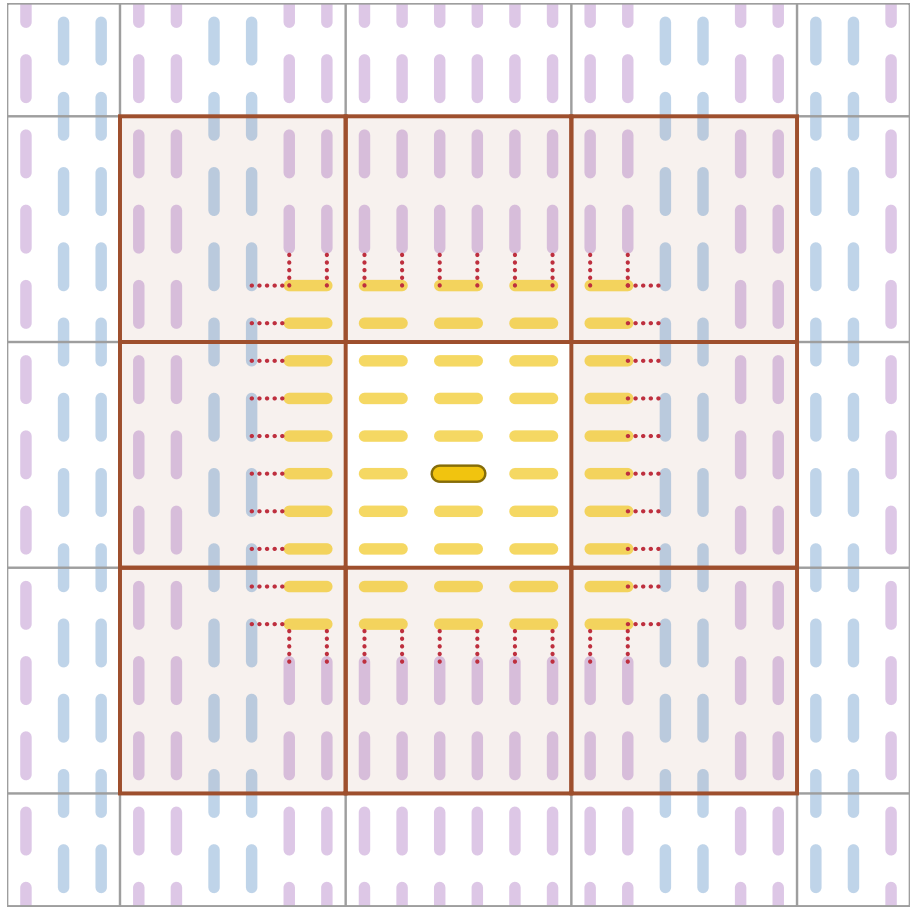
Proof strategy: Directly construct a $Ce^{-3\beta(1-1/\mathfrak{n})a}$-rare random set $B \subset \mathbb{Z}^2$ s.t. Θ_{ver} contains the unique infinite component of $\mathbb{Z}^2 \setminus B$ a.s.

Case 1: $\mathbb{Z}^2 \setminus \Theta_{\mathrm{ver}}$ picks out a small component of defects. Locate six broken links surrounding the component, yielding $Ce^{-3\beta a}$ for the block.



Step 2: Microscopic rarity of misaligned dimers (3/3)

Case 2: $\mathbb{Z}^2 \setminus \Theta_{\mathrm{ver}}$ picks out a large component of defects amid a *sea* of vertical dimers. Look at those blocks intersecting the boundary of the component. When $\mathfrak{n}$ is large, each block intersects a long chain of defects, yielding $Ce^{-3\beta(1-1/\mathfrak{n})a}$ per block.



Step 3: Disagreement percolation

Goal: Prove that there is no hidden positional order.

Theorem (Disagreement percolation)

Let $\mathbb{P}_{\mathrm{ver}}, \mathbb{P}'_{\mathrm{ver}}$ be $\mathbb{Z}^2$-ergodic Gibbs measures under each of which Ψ_{ver} percolates. Sample $\sigma \sim \mathbb{P}_{\mathrm{ver}}$ and $\sigma' \sim \mathbb{P}'_{\mathrm{ver}}$. If

$$(\mathbb{P}_{\mathrm{ver}} \otimes \mathbb{P}'_{\mathrm{ver}})(\{\sigma \neq \sigma'\} \text{ has no infinite component}) = 1,$$

then $\mathbb{P}_{\mathrm{ver}} = \mathbb{P}'_{\mathrm{ver}}$, and $\mathbb{P}_{\mathrm{ver}}$ is extremal.

Use $3a > \mu$ to select a mesoscopic length scale $\mathfrak{s}$ s.t.

$$e^{\frac{1}{2}\beta(\mu+3a)} \ll \mathfrak{s} \ll e^{3\beta(1-1/\mathfrak{n})a}.$$

On this length scale:

1. Horizontal dimers remain rare by Step 2, which confines disagreement components to 1D columns.
2. Vacancies arise with high probability, which deterministically terminate 1D disagreement paths.

Acknowledgments

I thank Ian Jauslin and Ron Peled for introducing me to this problem and for helpful discussions, and Joel Lebowitz for generous summer support. I also thank an anonymous referee for suggesting the simplified proof strategy presented here.

Questions, comments, and suggestions?